

1. NAME OF THE MEDICINAL PRODUCT

Telmisartan Tablets 40mg
Telmisartan Tablets 80mg

TRADE NAME OF THE PRODUCT

AURO TELMISARTAN 40
AURO TELMISARTAN 80

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

AURO TELMISARTAN 40mg Tablets
Each Uncoated Tablet Contains
Telmisartan Ph. Eur. 40mg

AURO TELMISARTAN 80mg Tablets
Each Uncoated Tablet Contains
Telmisartan Ph. Eur. 80mg

For the full list of excipients, see Section 6.1

3. PHARMACEUTICAL FORM

AURO TELMISARTAN 40

White to off-white, oblong shaped, biconvex uncoated tablets, debossed with 'N & 40' on either side of the score line on one side and plain on the other side.

AURO TELMISARTAN 80

White to off-white, oblong shaped, biconvex uncoated tablets, debossed with 'N & 80' on either side of the score line on one side and plain on the other side.

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

Hypertension

Treatment of essential hypertension.

Cardiovascular risk reduction

Telmisartan is indicated for reduction of the risk of myocardial infarction, stroke, or death from cardiovascular causes in patients 55 years of age or older at high risk of developing major cardiovascular events who are unable to take ACE inhibitors.

High risk of cardiovascular events can be evidenced by history of coronary artery disease, peripheral arterial disease, stroke, transient ischemic attack, or high-risk diabetes (insulin-dependent or non-insulin dependent) with evidence of end-organ damage. Telmisartan can be used in addition to other needed treatment (such as antihypertensive, antiplatelet or lipid-lowering therapy).

Studies of Telmisartan in this setting do not exclude that it may not preserve a meaningful fraction of the effect of the ACE inhibitor to which it was compared. Consider using the ACE inhibitor first, and, if it is stopped for cough only, consider re-trying the ACE inhibitor after the cough resolves.

Use of telmisartan with an ACE inhibitor is not recommended.

4.2 Dosage and Administration

Adults

Treatment of essential hypertension

The recommended dose is 40 mg once daily. In cases where the target blood pressure is not achieved, telmisartan dose can be increased to a maximum of 80 mg once daily. Alternatively, telmisartan may be used in combination with thiazide-type diuretics such as hydrochlorothiazide, which has been shown to have an additive blood pressure lowering effect with telmisartan. When considering raising the dose, it must be borne in mind that the maximum antihypertensive effect is generally attained four - eight weeks after the start of treatment.

In patients with severe hypertension treatment with telmisartan at doses up to 160 mg alone and in combination with hydrochlorothiazide 12.5 - 25 mg daily was well tolerated and effective.

Cardiovascular Risk Reduction

The recommended dose of AURO TELMISARTAN is 80mg once a day and can be administered with or without food. It is not known whether doses lower than 80mg of Telmisartan are effective in reducing the risk of cardiovascular morbidity and mortality.

When initiating AURO TELMISARTAN therapy for cardiovascular risk reduction, monitoring of blood pressure is recommended, and if appropriate adjustment of medications that lower blood pressure may be necessary.

Telmisartan tablets are for once-daily oral administration and should be taken with liquid, with or without food.

Route of administration: Oral

Renal Impairment

No posology adjustment is required for patients with renal impairment, including those on haemodialysis.

Telmisartan is not removed from blood by hemofiltration.

Hepatic Impairment

In patients with mild to moderate hepatic impairment the posology should not exceed 40 mg once daily.

Elderly

No dosing adjustment is necessary.

Children and Adolescents

The safety and efficacy of AURO TELMISARTAN for use in children below 18 years have not been established.

4.3 Contraindications

- Hypersensitivity to the active substance or to any of the excipients listed.
- Second and third trimester of pregnancy
- Lactation
- Biliary obstructive disorders.
- Severe hepatic impairment.
- The concomitant use of Telmisartan with Aliskiren-containing products is contraindicated in patients with diabetes mellitus or renal impairment (GFR < 60 mL/min/1.73 m²).

4.4 Special Warnings and Precautions for Use

Pregnancy

Angiotensin II receptor antagonists should not be initiated during pregnancy. Unless continued angiotensin II receptor antagonist therapy is considered essential, patients planning pregnancy should be changed to alternative antihypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with angiotensin II receptor antagonists should be stopped immediately, and, if appropriate, alternative therapy should be started (see Sections 4.3 and 4.6)

Hepatic Impairment

Telmisartan is not to be given to patients with cholestasis, biliary obstructive disorders or severe hepatic impairment (see Section 4.3) since Telmisartan is mostly eliminated with the bile. These patients can be expected to have reduced hepatic clearance for Telmisartan. Telmisartan should be used only with caution in patients with mild to moderate hepatic impairment.

Renovascular Hypertension

There is an increased risk of severe hypotension and renal insufficiency when patients with bilateral renal artery stenosis or stenosis of the artery to a single functioning kidney are treated with medicinal products that affect the renin-angiotensin-aldosterone system.

Renal Impairment and Kidney Transplantation

When Telmisartan is used in patients with impaired renal function, periodic monitoring of potassium and creatinine serum levels is recommended. There is no experience regarding the administration of Telmisartan in patients with recent kidney transplantation.

Intravascular Hypovolaemia

Symptomatic hypotension, especially after the first dose of Telmisartan, may occur in patients who are volume and/or sodium depleted by vigorous diuretic therapy, dietary salt restriction, diarrhoea or vomiting. Such conditions should be corrected before the administration of Telmisartan. Volume and/or sodium depletion should be corrected prior to administration of Telmisartan.

Dual Blockade of the Renin-Angiotensin-Aldosterone System (RAAS)

There is evidence that the concomitant use of ACE-inhibitors, angiotensin II receptor blockers or Aliskiren increases the risk of hypotension, hyperkalaemia and decreased renal function (including acute renal failure).

Dual blockade of RAAS through the combined use of ACE-inhibitors, angiotensin II receptor blockers or Aliskiren is therefore not recommended (see Sections 4.5 and 5.1)

If dual blockade therapy is considered absolutely necessary, this should only occur under specialist supervision and subject to frequent close monitoring of renal function, electrolytes and blood pressure.

ACE-inhibitors and angiotensin II receptor blockers should not be used concomitantly in patients with diabetic nephropathy.

Other Conditions with Stimulation of the Renin-Angiotensin-Aldosterone System

In patients whose vascular tone and renal function depend predominantly on the activity of the renin-angiotensin-aldosterone system (e.g. patients with severe congestive heart failure or underlying renal disease, including renal artery stenosis), treatment with medicinal products that affect this system such as Telmisartan has been associated with acute hypotension, hyperazotaemia, oliguria, or rarely acute renal failure (see Section 4.8)

Primary Aldosteronism

Patients with primary aldosteronism generally will not respond to antihypertensive medicinal products acting through inhibition of the renin-angiotensin system. Therefore, the use of telmisartan is not recommended.

Aortic and Mitral Valve Stenosis, Obstructive Hypertrophic Cardiomyopathy

As with other vasodilators, special caution is indicated in patients suffering from aortic or mitral stenosis, or obstructive hypertrophic cardiomyopathy.

Diabetic Patients Treated with Insulin or Antidiabetics

In these patients, hypoglycaemia may occur under telmisartan treatment. Therefore, in these patients an appropriate blood glucose monitoring should be considered; a dose adjustment of insulin or antidiabetics may be required, when indicated.

Hyperkalaemia

The use of medicinal products that affect the renin-angiotensin-aldosterone system may cause hyperkalaemia.

In the elderly, in patients with renal insufficiency, in diabetic patients, in patients concomitantly treated with other medicinal products that may increase potassium levels, and/or in patients with intercurrent events, hyperkalaemia maybe fatal.

Before considering the concomitant use of medicinal products that affect the renin-angiotensin-aldosterone system, the benefit risk ratio should be evaluated.

The main risk factors for hyperkalaemia to be considered are:

- Diabetes mellitus, renal impairment, age (> 70 years)
- Combination with one or more other medicinal products that affect the renin-angiotensin-aldosterone system and/or potassium supplements. Medicinal products or therapeutic classes of medicinal products that may provoke hyperkalaemia are salt substitutes containing potassium, potassium-sparing diuretics, ACE inhibitors, angiotensin II receptor antagonists, non-steroidal anti-inflammatory medicinal products (NSAIDs, including selective COX-2 inhibitors), heparin, immunosuppressives (cyclosporin or tacrolimus), and trimethoprim.
- Intercurrent events, in particular dehydration, acute cardiac decompensation, metabolic acidosis, worsening of renal function, sudden worsening of the renal condition (e.g., infectious diseases), cellular lysis (e.g. acute limb ischaemia, rhabdomyolysis, extend trauma).

Close monitoring of serum potassium in at risk patients is recommended (see Section 4.5).

Ethnic Differences

As with other angiotensin converting enzyme inhibitors, telmisartan and the other angiotensin II receptor antagonists are apparently less effective in lowering blood pressure in black people than in non-blacks, possibly because of higher prevalence of low-renin states in the black hypertensive population.

Other

As with any antihypertensive agent, excessive reduction of blood pressure in patients with ischaemic cardiopathy or ischaemic cardiovascular disease could result in a myocardial infarction or stroke.

4.5 Interactions with Other Medicinal Products and Other Forms of Interaction

Digoxin

When telmisartan is co-administered with digoxin, there are median increases in digoxin peak plasma concentration (49%) and in trough concentration (20%). When initiating, adjusting, and discontinuing telmisartan, monitor digoxin levels in order to maintain levels within the therapeutic range.

As with other medicinal products acting on the renin-angiotensin-aldosterone system, telmisartan may provoke hyperkalaemia. The risk may increase in case of treatment combination with other medicinal products that may also provoke hyperkalaemia (salt substitutes containing potassium, potassium-sparing diuretics, ACE inhibitors, angiotensin II receptor antagonists, non-steroidal anti-inflammatory medicinal products (NSAIDs, including selective COX-2 inhibitors), heparin, immunosuppressives (cyclosporin or tacrolimus), and trimethoprim).

The occurrence of hyperkalaemia depends on associated risk factors. The risk is increased in case of the above-mentioned treatment combinations. The risk is particularly high in combination with potassium sparing-diuretics, and when combined with salt substitutes containing potassium. A combination with ACE inhibitors or NSAIDs, for example, presents a lesser risk provided that precautions for use are strictly followed.

Concomitant Use Not Recommended

Potassium Sparing Diuretics or Potassium Supplements

Angiotensin II receptor antagonists such as telmisartan, attenuate diuretic induced potassium loss. Potassium sparing diuretics e.g., spirinolactone, eplerenone, triamterene, or amiloride, potassium supplements, or potassium-containing salt substitutes may lead to a significant increase in serum potassium. If concomitant use is indicated because of documented hypokalaemia, they should be used with caution and with frequent monitoring of serum potassium.

Lithium

Reversible increases in serum lithium concentrations and toxicity have been reported during concomitant administration of lithium with angiotensin converting enzyme inhibitors, and with angiotensin II receptor antagonists, including telmisartan. If use of the combination proves necessary, careful monitoring of serum lithium levels is recommended.

Concomitant Use Requiring Caution

Non-Steroidal Anti-Inflammatory Medicinal Products

NSAIDs (i.e., acetylsalicylic acid at anti-inflammatory dosage regimens, COX-2 inhibitors and non-selective NSAIDs) may reduce the antihypertensive effect of angiotensin II receptor antagonists. In some patients with compromised renal function (e.g., dehydrated patients or elderly patients with compromised renal function), the co-administration of angiotensin II receptor antagonists and agents that inhibit cyclo-oxygenase may result in further deterioration of renal function, including possible acute renal failure, which is usually reversible. Therefore, the combination should be administered with caution, especially in the elderly. Patients should be adequately hydrated, and consideration should be given to monitoring of renal function after initiation of concomitant therapy and periodically thereafter.

Co-administration of telmisartan and ramipril leads to an increase of up to 2.5-fold in the AUC₀₋₂₄ and C_{max} of ramipril and ramiprilat. The clinical relevance of this observation is not known.

Diuretics (Thiazide or Loop Diuretics)

Prior treatment with high dose diuretics such as furosemide (loop diuretic) and hydrochlorthiazide (thiazide diuretic) may result in volume depletion, and in a risk of hypotension when initiating therapy with telmisartan.

To Be Taken Into Account With Concomitant Use

Other Antihypertensive Agents

The blood pressure lowering effect of telmisartan can be increased by concomitant use of other antihypertensive medicinal products.

Dual blockade of the renin-angiotensin-aldosterone-system (RAAS) through the combined use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren is associated with a higher frequency of adverse events such as hypotension, hyperkalaemia and decreased renal function (including acute renal failure) compared to the use of a single RAAS-acting agent.

Based on their pharmacological properties it can be expected that the following medicinal products may potentiate the hypotensive effects of all antihypertensives including telmisartan: Baclofen, amifostine. Furthermore, orthostatic hypotension may be aggravated by alcohol, barbiturates, narcotics, or antidepressants.

Corticosteroids (Systemic Use)

Reduction of the antihypertensive effect.

A/s: 210 x 360 mm ■ Black

 AUROBINDO Packaging Development		Product Name	Component	Item Code	Date & Time
		AURO TELMISARTAN Tablets	Leaflet	P1538104	22.10.2024 & 10.20 am
		Country	Version No.	Reason Of Issue	Reviewed / Approved by
		Malaysia_U7	00	Submission	
Team Leader	Kiran	Dimensions (mm)	Colours: 01		
Initiator	Shirisha	A/s : 210 x 360 mm	 38104		
Artist	Advnt (Kiran)	Pharma Code: 38104			
Additional Information: Supersede code: P1534057					

4.6 Fertility, Pregnancy and Lactation

Pregnancy

The use of angiotensin II receptor antagonists is not recommended during the first trimester of pregnancy. The use of angiotensin II receptor antagonists is contraindicated during the second and third trimesters of pregnancy.

There are no adequate data from the use of telmisartan in pregnant women. Reproductive toxicity has been seen in animals.

Epidemiological evidence regarding the risk of teratogenicity following exposure to ACE inhibitors during the first trimester of pregnancy has not been conclusive; however a small increase in risk cannot be excluded. Whilst there is no controlled epidemiological data on the risk with angiotensin II receptor antagonists, similar risks may exist for this class of medicinal products. Unless continued angiotensin II receptor antagonist therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with angiotensin II receptor antagonists should be stopped immediately, and, if appropriate, alternative therapy should be started.

Exposure to angiotensin II receptor antagonists therapy during the second and third trimesters is known to induce human foetotoxicity (decreased renal function, oligohydramnios, skull ossification retardation) and neonatal toxicity (renal failure, hypotension, hyperkalaemia).

Should exposure to angiotensin II receptor antagonists have occurred from the second trimester of pregnancy, ultrasound check of renal function and skull is recommended.

Infants whose mothers have taken angiotensin II receptor antagonists should be closely observed for hypotension.

Breast-Feeding

Because no information is available regarding the use of telmisartan during breast-feeding, telmisartan is not recommended and alternative treatments with better established safety profiles during breast-feeding are preferable, especially while nursing a newborn or preterm infant.

Fertility

No effects of telmisartan on male and female fertility were observed.

4.6 Effects on Ability to Drive and Use Machines

When driving vehicles or operating machinery it should be taken into account that dizziness or drowsiness may occasionally occur when taking antihypertensive therapy such as telmisartan.

4.8 Undesirable Effects

The safety profile of telmisartan for the reduction of cardiovascular morbidity was consistent with that for hypertension.

Tabulated Summary of Adverse Reactions

Very common	Common	Uncommon	Rare	Very rare
Infections and infestations				
		Upper respiratory tract infection including pharyngitis and sinusitis, urinary tract infection including cystitis	Sepsis including fatal outcome	
Blood and the lymphatic system disorders				
		Anaemia	Eosinophilia, thrombocytopenia	
Immune system disorders				
			Anaphylactic reaction, hypersensitivity	
Metabolism and nutrition disorders				
		Hyperkalaemia	Hypoglycaemia (in diabetic patients)	
Psychiatric disorders				
		Depression, insomnia	Anxiety	
Nervous system disorders				
		Syncope	Somnolence	
Eye disorders				
			Visual disturbance	
Ear and labyrinth disorders				
		Vertigo		
Cardiac disorders				
		Bradycardia	Tachycardia	
Vascular disorders				
		Hypotension, orthostatic hypotension		
Respiratory, thoracic and mediastinal disorders				
		Dyspnoea, cough	Interstitial lung disease	
Gastrointestinal disorders				
		Abdominal pain, diarrhoea, dyspepsia, flatulence, vomit	Stomach discomfort, dry mouth, dysgeusia	
Hepato-biliary disorders				
			Hepatic function abnormal/liver disorder	
Skin and subcutaneous tissue disorders				
		Hyperhidrosis, pruritus, rash	Angioedema (also with fatal outcome), eczema, erythema, urticaria, drug eruption, toxic skin eruption	
Musculoskeletal and connective tissue disorders				
		Myalgia, back pain (e.g., sciatica), muscle spasms	Arthralgia, pain in extremity, tendon pain (tendonitis like symptoms)	
Renal and urinary disorders				
		Renal impairment including acute renal failure		
General disorders and administration site conditions				
		Chest pain, asthenia (weakness)	Influenza-like illness	
Investigations				
		Blood creatinine increased	Blood uric acid increased, hepatic enzyme increased, blood creatine phosphokinase increased, haemoglobin decreased	

4.9 Overdose

There is limited information available with regard to overdose in humans.

Symptoms

The most prominent manifestations of telmisartan overdose were hypotension and tachycardia; bradycardia dizziness, increase in serum creatinine, and acute renal failure have also been reported.

Management

Telmisartan is not removed by haemodialysis. The patient should be closely monitored, and the treatment should be symptomatic and supportive. Management depends on the time since ingestion and the severity of the symptoms. Suggested measures include induction of emesis and/ or gastric lavage. Activated charcoal may be useful in the treatment of overdosage.

Serum electrolytes and creatinine should be monitored frequently. If hypotension occurs, the patient should be placed in a supine position, with salt and volume replacement given quickly.

5.0 PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Pharmacotherapeutic group: Angiotensin II Antagonists, plain, ATC Code: C09CA07.

Mechanism of Action

Telmisartan is an orally active and specific angiotensin II receptor (type AT₁) antagonist. Telmisartan displaces angiotensin II with very high affinity from its binding site at the AT₁ receptor subtype, which is responsible for the known actions of angiotensin II. Telmisartan does not exhibit any partial agonist activity at the AT₁ receptor. Telmisartan selectively binds the AT₁ receptor. The binding is long-lasting. Telmisartan does not show affinity for other receptors, including AT₂ and other less characterised AT receptors. The functional role of these receptors is not known, nor is the effect of their possible overstimulation by angiotensin II, whose levels are increased by telmisartan. Plasma aldosterone levels are decreased by telmisartan. Telmisartan does not inhibit human plasma renin or block ion channels. Telmisartan does not inhibit angiotensin converting enzyme (kininase II), the enzyme which also degrades bradykinin. Therefore, it is not expected to potentiate bradykinin-mediated adverse events.

In human, an 80 mg dose of telmisartan almost completely inhibits the angiotensin II evoked blood pressure increase. The inhibitory effect is maintained over 24 hours and still measurable up to 48 hours.

Treatment of Essential Hypertension

After the first dose of telmisartan, the antihypertensive activity gradually becomes evident within 3 hours. The maximum reduction in blood pressure is generally attained 4 weeks after the start of treatment and is sustained during long-term therapy. The antihypertensive effect persists constantly over 24 hours after dosing and includes the last 4 hours before the next dose as shown by ambulatory blood pressure measurements. This is confirmed by trough to peak ratios consistently above 80 % seen after doses of 40 and 80 mg of telmisartan.

In patients with hypertension telmisartan reduces both systolic and diastolic blood pressure without affecting pulse rate. The antihypertensive efficacy of telmisartan has been compared to antihypertensive drugs such as amlodipine, atenolol, enalapril, hydrochlorothiazide, losartan, lisinopril, ramipril and valsartan.

Upon abrupt cessation of treatment with telmisartan, blood pressure gradually returns to pre-treatment values over a period of several days without evidence of rebound hypertension. Telmisartan treatment is associated with significant reductions in Left Ventricular Mass and Left Ventricular Mass Index in patients with hypertension and Left Ventricular Hypertrophy.

Telmisartan treatment is associated with statistically significant reductions in proteinuria (including microalbuminuria and macroalbuminuria) in patients with hypertension and diabetic nephropathy. The incidence of dry cough is lower in patients given telmisartan compared to angiotensin converting enzyme inhibitors.

5.2 Pharmacokinetic Properties

Absorption of telmisartan is rapid although the amount absorbed varies. The mean absolute bioavailability for telmisartan is about 50%. When telmisartan is taken with food, the reduction in the area under the plasma concentration-time curve (AUC) of telmisartan varies from approximately 6% (40 mg dose) to approximately 19% (160 mg dose). By 3 hours after administration plasma concentrations are similar whether telmisartan is taken fasting or with food.

The small reduction in AUC is not expected to cause a reduction in the therapeutic efficacy.

C_{max} and AUC being approximately 3-and 2-fold higher, respectively, in females compared to males without relevant influence on efficacy.

Telmisartan is largely bound to plasma protein (> 99.5 %), mainly albumin and alpha-1 acid glycoprotein. The mean steady state apparent volume of distribution (V_{ss}) is approximately 500 L. Telmisartan is metabolised by conjugation to the glucuronide of the parent compound. No pharmacological activity has been shown for the conjugate.

Telmisartan is characterised by biexponential decay pharmacokinetics with a terminal elimination half-life of >20 hours. The maximum plasma concentration (C_{max}) and, to a smaller extent, area under the plasma concentration-time curve (AUC) increase disproportionately with dose. There is no evidence of clinically relevant accumulation of telmisartan.

Telmisartan is nearly exclusively excreted with the faeces, exclusively as unchanged compound. Cumulative urinary excretion is < 2% of dose. Total plasma clearance (CL_{tot}) is high (approximately 900 ml/min compared with hepatic blood flow (about 1500 ml/min).

Elderly Patients

The pharmacokinetics of telmisartan do not differ between younger and elderly patients.

Patients with Renal Impairment

Lower plasma concentrations were observed in patients with renal insufficiency undergoing dialysis. Telmisartan is highly bound to plasma protein in renal-insufficiency and cannot be removed by dialysis. The elimination half-life is not changed in patients with renal impairment.

Patients with Hepatic Impairment

Hepatic impairment showed an increase in absolute bioavailability up to nearly 100%. The elimination half-life is not changed in patients with hepatic impairment.

6. PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Mannitol, Meglumine, Povidone, Sodium Hydroxide, Colloidal Anhydrous Silica, Sodium Stearyl Fumarate.

6.2 Shelf Life

36 months from date of manufacturing (please refer to outer package for expiry date)

6.3 Special Precautions for Storage

Store in a dry place below 30°C. Store in original package in order to protect from moisture.

6.4 Nature and Contents of Container

10 tablets per blister, 3 blisters per box.

7.0 MANUFACTURER AND PRODUCT REGISTRATION HOLDER

7.1 Manufacturer

Aurobindo Pharma Limited (Unit VII)
Special Economic Zone, TSIIIC, Plot No. S1, Sy. Nos., 411/P, 425/P, 434/P, 435/P and 458/P,
Green Industrial Park, Polepally Village, Jadcherla Mandal,
Mahaboobnagar District, Telangana State, 509302, India

7.2 Product Registration Holder

Synervv Sdn. Bhd.,SO-29-2, Menara 1, KL Eco City, Jalan Bangsar, Kg Haji Abdullah Hukum, 59200, Kuala Lumpur, Malaysia

8.0 PACKAGE INSERT REVISION DATE

October 2024